4ª Lista de Exercícios de Cálculo I

INTEGRAL INDEFINIDA

1. Calcule

a) $\int x\,dx$

b) $\int 3\,dx$

c) $\int (3x+1)\,dx$

d) $\int (x^2+x+1)\,dx$

e) $\int x^3\,dx$

f) $\int (x^3+2x+3)\,dx$

g) $\int \frac{1}{x^2}\,dx$

h) $\int \left(x+\frac{1}{x^3}\right)dx$

i) $\int \sqrt{x}\,dx$

j) $\int \sqrt[3]{x}\,dx$

l) $\int \left(x+\frac{1}{x}\right)dx$

m) $\int (2+\sqrt[4]{x})\,dx$

n) $\int (ax+b)\,dx$, a e b constantes

o) $\int \left(3x^2+x+\frac{1}{x^3}\right)dx$

p) $\int \left(\sqrt{x}+\frac{1}{x^2}\right)dx$

q) $\int \left(\frac{2}{x}+\frac{3}{x^2}\right)dx$

r) $\int (3\sqrt[3]{x^2}+3)\,dx$

s) $\int \left(2x^3-\frac{1}{x^4}\right)dx$

t) $\int \frac{x^2+1}{x}\,dx$

2. Calcule

a) $\int e^{2x}\,dx$

b) $\int e^{-x}\,dx$

c) $\int (x+3e^x)\,dx$

d) $\int \cos 3x\,dx$

e) $\int \operatorname{sen} 5x\,dx$

f) $\int (e^{2x}+e^{-2x})\,dx$

g) $\int (x^2+\operatorname{sen} x)\,dx$

h) $\int (3+\cos x)\,dx$

i) $\int \frac{e^x+e^{-x}}{2}\,dx$

j) $\int \frac{1}{e^{3x}}\,dx$

l) $\int (\operatorname{sen} 3x+\cos 5x)\,dx$

m) $\int \left(\frac{1}{x}+e^x\right)dx,\ x>0$

n) $\int \operatorname{sen}\frac{x}{2}\,dx$

o) $\int \cos\frac{x}{3}\,dx$

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INTEGRAL DE RIEMANN

3. Calcule

1. $\int_0^1 (x+3)\,dx$
2. $\int_{-1}^1 (2x+1)\,dx$
3. $\int_0^4 \frac{1}{2}\,dx$
4. $\int_{-2}^1 (x^2-1)\,dx$
5. $\int_1^3 dx$
6. $\int_{-1}^2 4\,dx$
7. $\int_1^3 \frac{1}{x^3}$
8. $\int_{-1}^1 5\,dx$
9. $\int_0^2 (x^2+3x-3)\,dx$
10. $\int_0^1 \left(5x^3-\frac{1}{2}\right)dx$
11. $\int_1^1 (2x+3)\,dx$
12. $\int_1^0 (2x+3)\,dx$
13. $\int_{-2}^{-1} \left(\frac{1}{x^2}+x\right)dx$
14. $\int_0^4 \sqrt{x}$
15. $\int_1^4 \frac{1}{\sqrt{x}}\,dx$
16. $\int_0^8 \sqrt[3]{x}\,dx$
17. $\int_{-1}^0 (x^3-2x+3)\,dx$
18. $\int_0^1 \sqrt[8]{x}\,dx$
19. $\int_1^2 \left(x^3+x+\frac{1}{x^3}\right)dx$
20. $\int_0^1 (x+\sqrt[4]{x})\,dx$
21. $\int_1^3 \left(5+\frac{1}{x^2}\right)dx$
22. $\int_{-3}^3 x^3\,dx$
23. $\int_{-1}^1 (x^7+x^3+x)\,dx$
24. $\int_{\frac{1}{2}}^1 (x+3)\,dx$
25. $\int_1^4 (5x+\sqrt{x})\,dx$
26. $\int_1^0 (x^7-x+3)\,dx$
27. $\int_1^2 \frac{1+x}{x^3}\,dx$
28. $\int_0^1 (x+1)^2\,dx$
29. $\int_1^4 \frac{1+x}{\sqrt{x}}\,dx$
30. $\int_0^1 (x-3)^2\,dx$
31. $\int_0^2 (t^2+3t-1)\,dt$
32. $\int_1^2 \frac{1+t^2}{t^4}\,dt$
33. $\int_{\frac{1}{2}}^1 (s+2)\,ds$
34. $\int_0^3 (u^2-2u+3)\,du$
35. $\int_1^2 (s^2+3s+1)\,ds$
36. $\int_{-1}^1 \sqrt[3]{t}\,dt$
37. $\int_1^3 \left(1+\frac{1}{x}\right)dx$
38. $\int_1^2 \frac{1+3x^2}{x}\,dx$
39. $\int_{-\frac{\pi}{3}}^{\frac{\pi}{2}} \cos 2x\,dx$
40. $\int_{-\pi}^0 \operatorname{sen} 3x\,dx$
42. $\int_0^1 \frac{1}{1+t^2}\,dt$
43. $\int_0^{\frac{\pi}{4}} \operatorname{sen} x\,dx$
44. $\int_{-1}^0 e^{-2x}\,dx$
45. $\int_0^{\frac{\pi}{3}} (3+\cos 3x)\,dx$
46. $\int_0^1 \operatorname{sen} 5x\,dx$

CÁLCULO DE ÁREAS

Nos exercícios abaixo, calcule a área do conjunto A dado:

1) A é o conjunto do plano limitado pelas retas $x = 1, x = 3$, pelo $eixo\ x$ e pelo gráfico de $y = x^3$.
2) A é o conjunto de todos (x,y) tais que $x^2 - 1 \leq y \leq 0$.
3) A é o conjunto limitado pelas retas $x = -1, x = 2, y = 0$ e pelo gráfico de $y = x^2 + 2x + 5$.
4) A é o conjunto de todos (x,y) tais que $x \geq 0$ e $x^3 \leq y \leq x$.
5) A é o conjunto de todos os pontos (x,y) tais que $x^2 - 1 \leq y \leq x + 1$.
6) A é o conjunto do plano limitado pelas retas $x = 0, x = \frac{\pi}{2}$ e pelos gráficos de $y = senx$ e $y = cosx$.

MUDANÇA DE VARIÁVEL

4. Calcule

a) $\int_1^2 (x-2)^5\,dx$

b) $\int_0^1 (3x+1)^4\,dx$

c) $\int_0^1 \sqrt{3x+1}\,dx$

d) $\int_{-1}^0 (2x+5)^3\,dx$

e) $\int_{-3}^4 \sqrt[3]{5-x}\,dx$

f) $\int_1^2 \frac{2}{(3x-2)^3}\,dx$

g) $\int_0^1 \frac{1}{(x+1)^5}\,dx$

h) $\int_{-2}^1 \frac{3}{4+x}\,dx$

i) $\int_0^2 e^{2x}\,dx$

j) $\int_0^1 xe^{x^2}\,dx$

l) $\int_{-1}^0 x\sqrt{x+1}\,dx$

m) $\int_0^{\frac{\pi}{3}} \cos 2x\,dx$

n) $\int_0^1 \frac{x^2}{1+x^3}\,dx$

o) $\int_0^1 \frac{x^2}{(1+x^3)^2}\,dx$

p) $\int_{-1}^0 x^2\sqrt{1+x^3}\,dx$

q) $\int_1^3 \frac{2}{5+3x}\,dx$

r) $\int_{-1}^1 \sqrt[3]{x+1}\,dx$

s) $\int_0^1 \frac{x}{(x+1)^5}\,dx$

t) $\int_{-1}^0 x(x+1)^{100}\,dx$

u) $\int_1^2 x^2(x-2)^{10}\,dx$

5. Calcule

a) $\int_0^1 x\sqrt{x^2+3}\,dx$

b) $\int_0^1 x(x^2+3)^5\,dx$

c) $\int_1^2 x(x^2-1)^5\,dx$

d) $\int_0^1 x\sqrt{1-x^2}\,dx$

e) $\int_{-1}^0 x^2e^{x^3}\,dx$

f) $\int_0^1 x\sqrt{1+2x^2}\,dx$

g) $\int_1^2 \frac{3s}{1+s^2}\,ds$

h) $\int_0^1 \frac{1}{1+4s}\,ds$

i) $\int_0^3 \frac{x}{\sqrt{x+1}}\,dx$

j) $\int_0^1 \frac{s}{\sqrt{s^2+1}}\,ds$

l) $\int_0^3 \frac{x^2}{\sqrt{x+1}}\,dx$

m) $\int_0^1 \frac{x^2}{(x+1)^2}\,dx$

n) $\int_{-1}^1 x^3(x^2+3)^{10}\,dx$

o) $\int_0^{\sqrt{3}} x^3\sqrt{x^2+1}\,dx$

p) $\int_0^{\frac{\pi}{3}} \operatorname{sen} x\cos^2 x\,dx$

q) $\int_0^{\frac{\pi}{6}} \cos x\operatorname{sen}^5 x\,dx$

r) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \operatorname{sen} x(1-\cos^2 x)\,dx$

INTEGRAÇÃO POR PARTES

6. Calcule

a) $\int x\,e^x\,dx$

b) $\int x\operatorname{sen} x\,dx$

c) $\int x^2\,e^x\,dx$

d) $\int x\ln x\,dx$

e) $\int \ln x\,dx$

f) $\int x^2\ln x\,dx$

g) $\int x\sec^2 x\,dx$

h) $\int x(\ln x)^2\,dx$

i) $\int (\ln x)^2\,dx$

j) $\int x\,e^{2x}\,dx$

l) $\int e^x\cos x\,dx$

m) $\int e^{-2x}\operatorname{sen} x\,dx$

n) $\int x^3\,e^{x^2}\,dx$

o) $\int x^3\cos x^2\,dx$

p) $\int e^{-x}\cos 2x\,dx$

q) $\int x^2\operatorname{sen} x\,dx$

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RESPOSTAS

1. *a)* $\frac{x^2}{2} + k$ *b)* $3x + k$ *c)* $\frac{3x^2}{2} + x + k$ *d)* $\frac{x^3}{3} + \frac{x^2}{2} + x + k$

e) $\frac{x^4}{4} + k$ *f)* $\frac{x^4}{4} + x^2 + 3x + k$ *g)* $-\frac{1}{x} + k$ *h)* $\frac{x^2}{2} - \frac{1}{2x^2}$

i) $\frac{2}{3}\sqrt{x^3} + k$ *j)* $\frac{3}{4}\sqrt[3]{x^4}$ *l)* $\frac{x^2}{2} + \ln x + k$ *m)* $2x + \frac{4}{5}\sqrt[4]{x^5}$

n) $\frac{a}{2}x^2 + bx + k$ *o)* $x^3 + \frac{x^2}{2} - \frac{1}{2x^2} + k$ *p)* $\frac{2}{3}\sqrt{x^3} - \frac{1}{x} + k$

q) $2\ln x - \frac{3}{x} + k$ *r)* $\frac{15}{7}\sqrt[5]{x^7} + 3x + k$

s) $\frac{x^4}{2} + \frac{1}{3x^3} + k$ *t)* $\frac{x^2}{2} + \ln x + k$

2. *a)* $\frac{1}{2}e^{2x} + k$ *b)* $-e^{-x} + k$ *c)* $\frac{x^2}{2} + 3e^x + k$ *d)* $\frac{1}{3}\text{sen } 3x + k$

e) $-\frac{1}{5}\cos 5x + k$ *f)* $\frac{1}{2}e^{2x} - \frac{1}{2}e^{-2x} + k$ *g)* $\frac{x^3}{3} - \cos x + k$

h) $3x + \text{sen } x + k$ *i)* $\frac{1}{2}(e^x - e^{-x}) + k$ *j)* $-\frac{1}{3}e^{-3x} + k$

l) $-\frac{1}{3}\cos 3x + \frac{1}{5}\text{sen } 5x + k$ *m)* $\ln x + e^x + k$ *n)* $-2\cos\frac{x}{2} + k$

3.

1. 7/2	**2.** 2	**3.** 2	**4.** 0	**5.** 2	**6.** 12	**7.** 4/9
8. 10	**9.** 8/3	**10.** 3/4	**11.** 0	**12.** −4	**13.**	
14. 16/3	**15.** 2	**16.** 12	**17.** 15/4	**18.** 8/9	**19.** 45/8	
20. 13/10	**21.** 32/3	**22.** 0	**23.** 0	**24.** 15/8	**25.** 253/6	
26. −21/8	**27.** 7/8	**28.** 7/3	**29.** 20/3	**30.** 19/3	**31.** 20/3	
32. 19/24	**33.** 11/8	**34.** 9	**35.** 47/6	**36.** 0	**37.** $2 + \ln 3$	

38. $\ln 2 + \frac{9}{2}$ **39.** $\frac{\sqrt{3}}{4}$ **40.** $-\frac{2}{3}$ **41.** $\frac{1}{2}(e^2 - e^{-2})$ **42.** $\frac{\pi}{4}$

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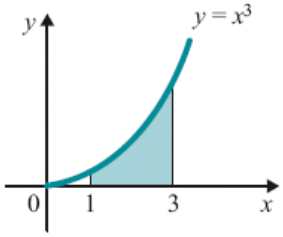
Cálculo de áreas

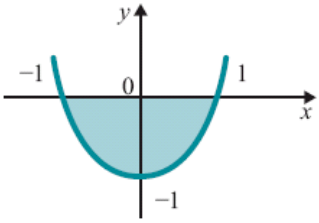
1)

Área = 20



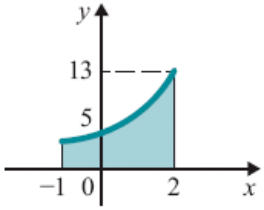
2)

Área = $\frac{4}{3}$



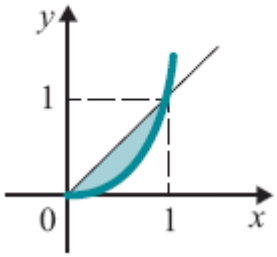
3)

Área = 21



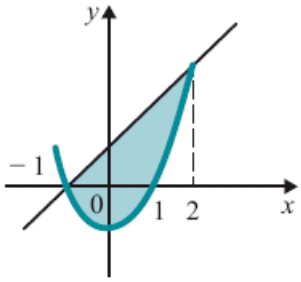
4)

Área = $\frac{1}{4}$



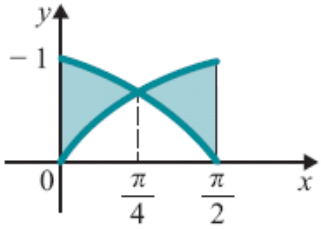
5)

Área = $\frac{9}{2}$



6)

Área = $2(\sqrt{2} - 1)$



MUDANÇA DE VARIÁVEL

4.

a) $-\frac{1}{6}$ *b)* $\frac{1.023}{15}$ *c)* $\frac{14}{9}$ *d)* 68 *e)* $\frac{45}{4}$ *f)* $\frac{5}{16}$

g) $\frac{15}{64}$ *h)* $3\ln\frac{5}{2}$ *i)* $\frac{1}{2}(e^4 - 1)$ *j)* $\frac{1}{2}(e - 1)$ *l)* $-\frac{4}{15}$

m) $\frac{\sqrt{3}}{4}$ *n)* $\frac{1}{3}\ln 2$ *o)* $\frac{1}{6}$ *p)* $\frac{2}{9}$ *q)* $\frac{2}{3}\ln\frac{7}{4}$ *r)* 0

s) $\frac{11}{192}$ *t)* $-\frac{1}{10.302}$ *u)* $\frac{46}{429}$

5.

a) $\dfrac{8-\sqrt{27}}{3}$ *b)* $\dfrac{3.367}{12}$ *c)* $\dfrac{243}{4}$ *d)* $\dfrac{1}{3}$ *e)* $\dfrac{1}{3}(1-e^{-1})$

f) $\dfrac{3\sqrt{3}-1}{6}$ *g)* $\dfrac{3}{2}\ln\dfrac{5}{2}$ *h)* $\dfrac{1}{4}\ln 5$ *i)* $\dfrac{8}{3}$ *j)*

l) $\dfrac{76}{15}$ *m)* $\dfrac{3-4\ln 2}{2}$ *n)* 0 *o)* $\dfrac{58}{15}$ *p)* $\dfrac{7}{24}$ *q)* $\dfrac{1}{384}$

r) $\dfrac{11}{24}$

6. *a)* $(x-1)e^x + k$ *b)* $-x\cos x + \operatorname{sen} x + k$ *c)* $e^x(x^2-2x+2)+k$

d) $\dfrac{x^2}{2}\left(\ln - \dfrac{1}{2}\right) + k$ *e)* $x(\ln x - 1) + k$ *f)* $\dfrac{1}{3}x^3\left(\ln x - \dfrac{1}{3}\right) + k$

g) $x\operatorname{tg} x + \ln|\cos x| + k$ *h)* $\dfrac{x^2}{2}\left[(\ln x)^2 - \ln x + \dfrac{1}{2}\right] + k$

i) $x(\ln x)^2 - 2x(\ln x - 1) + k$ *j)* $\dfrac{1}{2}e^{2x}\left(x - \dfrac{1}{2}\right) + k$

l) $\dfrac{1}{2}e^x(\operatorname{sen} x + \cos x) + k$ *m)* $-\dfrac{1}{5}e^{-2x}(\cos x + 2\operatorname{sen} x) + k$

n) $\dfrac{1}{2}(x^2-1)e^{x^2} + k$ *o)* $\dfrac{1}{2}(x^2\operatorname{sen} x^2 + \cos x^2) + k$

p) $\dfrac{e^{-x}}{5}(2\operatorname{sen} 2x - \cos 2x) + k$ *q)* $-2x^2\cos x + 2x\operatorname{sen} x + 2\cos x + k$